

Experimental Parameter / Specification	Configuration Setting
Model Architecture	Hybrid 1D-BiCNN-GRU
Trainable Parameters	257,541 (~257k) [Tuned from 578,309 baseline]
Parameter Compression Ratio	55.5% reduction via Optuna hyperparameter tuning
Input Signal Dimension	1D ECG (256 samples, centered R-peak)
Target Classes	5 Classes (AAMI EC57: N, SVEB, VEB, F, Q)
Loss Function	Categorical Cross-Entropy
Imbalance Handling	Class-Weighted Random Sampling
Sampling Exponent (α)	0.7320 [w_c = (1 / N_c)^0.7320]
Optimization Algorithm	AdamW Optimizer
Initial Learning Rate (η)	1.184×10^{-3} (0.001184)
Weight Decay	7.114×10^{-4} (0.000711)
Dropout Probability	0.2076
Mini-Batch Size	128
Training Epochs	20 epochs per fold
Evaluation Protocol	Patient-Independent 3-Fold Cross-Validation
Validation Benchmark	MIT-BIH Arrhythmia Database (AAMI DS1 Partition)